

③ LDE or NLDE

LDE form:

$$a_n(x)y^{(n)} + a_{n-1}(x)y^{(n-1)} + \dots + a_1(x)y' + a_0(x)y = f(x)$$

1. y and its derivative Degree of 1
2. ~~$y \times (y)'$ term~~
3. No nonlinear function of y

④ constant coefficient
vs.
variable coefficient
if $a_0, a_1, \dots, a_n$
all constant
one is not constant

$\frac{d}{dt} + 5(\frac{d}{dt})^2 + 3x = \sin t$
→ 4 order
(3) $\frac{\partial u}{\partial s} + \frac{\partial u}{\partial t} = 0 \rightarrow 1 \text{ order}$

2. separable ODEs modeling

(0) 直接積分

$$g(y)y' = f(x)$$

$$\int g(y)y' dx = \int f(x) dx + C$$

$$\rightarrow \int g(y) dy = \int f(x) dx + C$$

(1) 可分離變數 ① 可直接分離
② 變數變換 → ①
x放一邊, y放一邊

① $y' = f(x, y)$

$$\rightarrow \frac{dy}{dx} = f(x) \cdot f_2(y)$$

$$\rightarrow \int \frac{dy}{f_2(y)} = \int f(x) dx + C$$

$$\begin{aligned} M(x, y)dx + N(x, y)dy &= 0 \\ \rightarrow M_1(x)M_2(y)dx + N_1(x)N_2(y)dy &= 0 \\ \rightarrow \int \frac{M_1(x)}{N_1(x)} dx &= - \int \frac{N_2(y)}{M_2(y)} dy + C \\ \rightarrow \int \frac{M_1(x)}{N_1(x)} dx + \int \frac{N_2(y)}{M_2(y)} dy &= C \end{aligned}$$

② 可藉由變數變換寫成可分離變數ODE

(1) 壹階齊次ODE

(2) 型式為 $(a_1x + b_1y + c_1)dx + (a_2x + b_2y + c_2)dy = 0$

(3) 型式為 $y' = f(ax + by + c)$

△ 型式 $(a_1x + b_1y + c_1)dx + (a_2x + b_2y + c_2)dy = 0$

↳ $c_1 \neq 0$ or $c_2 \neq 0$ 且 $\frac{a_1}{a_2} = \frac{b_1}{b_2} = m \neq \frac{c_1}{c_2}$

↳ 令 $\begin{cases} a_2x + b_2y = z \\ a_1x + b_1y = mz \end{cases} \rightarrow a_2dx + b_2dy = dz, dy = \frac{dz - a_2dx}{b_2}$
△ 回原式: $(mz + c_1)dx + (z + c_2)(\frac{dz - a_2dx}{b_2}) = 0$

② - (1)

壹階齊次ODE

Def: 可改寫成 $y' = f(\frac{y}{x})$ or $y'(\frac{x}{y})$

① $y' = f(x, y)$

if $f(\lambda x, \lambda y) = \lambda^k f(x, y)$

② $M(x, y)dx + N(x, y)dy = 0$

$M(\lambda x, \lambda y) = \lambda^k \cdot M(x, y)$

$N(\lambda x, \lambda y) = \lambda^k \cdot N(x, y)$ k 隨意

solution:

1° 判斷是否為 $y' = f(\frac{y}{x})$ or $y' = f(\frac{x}{y})$ 型

2° 整理成 $y' = f(\frac{y}{x})$ or $y' = f(\frac{x}{y})$

3° 變數變換 $u = \frac{y}{x} \rightarrow y = ux$

$\rightarrow dy = xdu + udx$

代回原式

4° 分離變數 (回到①)

② - (2) $M(x, y)$

$N(x, y)$

型式 $(a_1x + b_1y + c_1)dx + (a_2x + b_2y + c_2)dy = 0$

① $c_1 = c_2 = 0$ (回到② - (1) 齊次ODE)

② $c_1 \neq 0$ or $c_2 \neq 0$ $\Delta. \frac{a_1}{a_2} \neq \frac{b_1}{b_2} \rightarrow$ 相交於一點之兩直線
 $\Delta. \frac{a_1}{a_2} = \frac{b_1}{b_2} = m \neq \frac{c_1}{c_2} \rightarrow$ 二條平行線

△ $\begin{cases} a_1x + b_1y + c_1 = 0 \rightarrow a_1x + b_1\beta + c_1 = 0 \\ a_2x + b_2y + c_2 = 0 \rightarrow a_2x + b_2\beta + c_2 = 0 \end{cases}$ 求其交集 $(X, y) = (\alpha, \beta)$

令 $\begin{cases} x = u + \alpha \\ y = v + \beta \end{cases}, dx = du$

$y = v + \beta, dy = dv$

變換 常數

代回原式:

$[a_1(u + \alpha) + b_1(v + \beta) + c_1]du + [a_2(u + \alpha) + b_2(v + \beta) + c_2]dv = 0$

$\Rightarrow (a_1u + b_1v)du + (a_2u + b_2v)dv = 0$

$\Rightarrow (a_1 + b_1\frac{v}{u})du + (a_2 + b_2\frac{v}{u})dv = 0$

令 $t = \frac{v}{u}$ (回到變數變換)

型 $y' = f(ax+by+c)$
 $t = ax+by+c$
 $dt = adx + bdy$
 $dy = \frac{dt - adx}{b}$
 代回原式

Ex: $(x^2y^2 + e^y) \frac{dy}{dx} + (xy^3 + 1) = 0$
 $\rightarrow \underbrace{2(xy^2+1)}_M dx + \underbrace{(x^2y^2+e^y)}_N dy = 0$
 $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} (2xy^2+2) = 6xy^2$
 $\frac{\partial N}{\partial x} = \frac{\partial}{\partial x} (x^2y^2+e^y) = 6xy^2$

$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \Rightarrow$ 全微分
 $M = 2xy^2 + 2 \Rightarrow \int 2xy^2 dx = x^2y^2 + 2x + f(y)$
 $N = x^2y^2 + e^y \Rightarrow \int x^2y^2 dy = \frac{1}{3}x^2y^3 + e^y + g(x)$
 $\therefore \phi(x,y) = x^2y^3 + 2x + e^y = C$

3. Exact ODE & Integrating factors

如何判 $Mdx + Ndy$ 正合 (Exact)?

$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \Leftrightarrow$ 正合

Solution:

$\phi(x,y) = C$

$M = \frac{\partial \phi}{\partial x}, N = \frac{\partial \phi}{\partial y}$

$I(x,y)$ 積分因子

常見四種 $I(x), I(y), I(x+y), I(xy)$

4. Linear ODE

型 $y' + p(x)y = q(x)$

$\frac{d}{dx} [y \cdot e^{\int p(x)dx}] = q(x) \cdot e^{\int p(x)dx}$

$\cos y \cdot y' = \sin(x+y)$

$\Rightarrow \cos y \cdot y' = \sin x \cos y + \cos x \sin y$

$\Rightarrow y' = \sin x + \cos x \tan y$

$\frac{1}{1+u^2} \frac{du}{dx} = \sin x + u \cos x$

$\frac{du}{dx} = (1+u^2)(\sin x + u \cos x)$

$\int \ln|x| dx$ $u = \ln x, dv = dx$
 $du = \frac{1}{x} dx, v = x$

$= x \ln x - \int x \cdot \frac{1}{x} dx$

$= x \ln x - x + C$

① $\frac{x^4 - 3x^2 + 1}{x(x-1)(x+1)(x+2)} = \frac{A}{x} + \frac{B}{x-1} + \frac{C}{x+1} + \frac{D}{x+2}$

$+ \frac{E}{x+3} + \frac{F}{x+4}$

不足正合 (即 $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$)

$\rightarrow$ 正合化

$I(x,y) [M(x,y)dx + N(x,y)dy] = 0 \cdot I(x,y)$

$\rightarrow I(x,y)$ 即積分因子 $\rightarrow$ 式子變正合

$\Rightarrow IMdx + INdy = 0$

$\frac{\partial(IM)}{\partial y} = \frac{\partial(IN)}{\partial x}$

$\frac{\partial}{\partial y} (I \frac{\partial M}{\partial y} + M \frac{\partial I}{\partial y}) = \frac{\partial}{\partial x} (I \frac{\partial N}{\partial x} + N \frac{\partial I}{\partial x})$

$\Rightarrow I (\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}) = N \frac{\partial I}{\partial x} - M \frac{\partial I}{\partial y}$

③ Assume $I(x,y) = I(x+y)$
 $I = I(x+y) = I(u) \quad u = x+y$

$\frac{(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x})}{N-M} \cdot dy = \frac{1}{I} dI$

$= f(x+y) \quad \uparrow \quad d(x+y)$

$\rightarrow \int f(x+y) d(x+y) = \ln|I|$

$\rightarrow I = e^{\int f(x+y) d(x+y)}$

④ $\frac{R(x)}{Q(x)} = \frac{R(x)}{(a_1x+b_1)(a_2x+b_2)\dots(a_nx+b_n)} \Rightarrow \frac{R(x)}{Q(x)} = \frac{A_1x+B_1}{a_1x+b_1} + \dots + \frac{A_nx+B_n}{a_nx+b_n}$

$(e^{ax})' = ae^{ax}$

$(a^x)' = a^x \ln a$

$\int x^n dx = \frac{x^{n+1}}{n+1} + C$

$\int \frac{1}{x} dx = \ln|x| + C$

$\int e^{ax} dx = \frac{1}{a} e^{ax} + C$

$\int \sin x dx = -\cos x + C$

$\int \cos x dx = \sin x + C$

$\int \tan x dx = -\ln|\cos x| + C$

$\int \cot x dx = \ln|\sin x| + C$

$\int \sec x dx = \ln|\sec x + \tan x| + C$

$\int \csc x dx = \ln|\csc x - \cot x| + C$

$(\sin^{-1}x)' = \frac{1}{\sqrt{1-x^2}}$

$(\cos^{-1}x)' = \frac{-1}{\sqrt{1-x^2}}$

$(\tan^{-1}x)' = \frac{1}{1+x^2}$

$(\cot^{-1}x)' = \frac{-1}{1+x^2}$

$(\sin x)' = \cos x$

$(\cos x)' = -\sin x$

$(\tan x)' = \sec^2 x$

$(\cot x)' = -\csc^2 x$

$(\sinh x)' = \cosh x$

$(\cosh x)' = \sinh x$

$(\ln x)' = \frac{1}{x}$

$(\log_a x)' = \frac{\log_a e}{x}$

$(\sin^{-1}x)' = \frac{1}{\sqrt{1-x^2}}$

$(\cos^{-1}x)' = \frac{-1}{\sqrt{1-x^2}}$

$(\tan^{-1}x)' = \frac{1}{1+x^2}$

$(\cot^{-1}x)' = \frac{-1}{1+x^2}$

$\int \frac{dx}{x^2+a^2} = \frac{1}{a} \tan^{-1} \frac{x}{a} + C$

$\int \frac{dx}{\sqrt{a^2-x^2}} = \sin^{-1}(\frac{x}{a}) + C$

① Assume $I(x,y) = I(x)$

$\frac{(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x})}{N} dx = \frac{1}{I} dI$

$\rightarrow \ln|I| = \int f(x) dx \rightarrow I = e^{\int f(x) dx}$

② Assume $I(x,y) = I(y)$

$\frac{(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x})}{-M} dy = \frac{1}{I} dI$

$\rightarrow \ln|I| = \int g(y) dy \rightarrow I = e^{\int g(y) dy}$

③ Assume $I(x,y) = I(xy)$ $I = I(xy) = I(u)$
 $u = xy$

$\frac{(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x})}{Ny - Mx} \cdot dy = \frac{1}{I} dI$

$= f(xy) \quad \uparrow \quad d(xy)$

$\rightarrow \int f(xy) d(xy) = \ln|I| \rightarrow I = e^{\int f(xy) d(xy)}$

$\int \frac{dx}{x^2+a^2} = \frac{1}{a} \tan^{-1} \frac{x}{a} + C$

$\int \frac{dx}{\sqrt{a^2-x^2}} = \sin^{-1}(\frac{x}{a}) + C$

$\int \frac{dx}{\sqrt{x^2+a^2}} = \sinh^{-1}(\frac{x}{a}) + C$

$\int \frac{dx}{\sqrt{x^2-a^2}} = \cosh^{-1}(\frac{x}{a}) + C$

$\int \sin^2 x dx = \frac{1}{2}x - \frac{1}{4}\sin 2x + C$

$\int \cos^2 x dx = \frac{1}{2}x + \frac{1}{4}\sin 2x + C$

$\int \tan^2 x dx = \tan x - x + C$

$\int \sec^2 x dx = \tan x + C$

$\int \csc^2 x dx = -\cot x + C$

$\int \frac{dx}{1+x^2} = \tan^{-1} x + C$

$\int \frac{dx}{1-x^2} = \frac{1}{2} \ln \left| \frac{1+x}{1-x} \right| + C$

$\int \frac{dx}{x^2-1} = \frac{1}{2} \ln \left| \frac{x-1}{x+1} \right| + C$

$\int \frac{dx}{x^2+1} = \tan^{-1} x + C$

$\int \frac{dx}{x^2-1} = \frac{1}{2} \ln \left| \frac{x-1}{x+1} \right| + C$

$\int \frac{dx}{x^2+1} = \tan^{-1} x + C$

$\int \frac{dx}{x^2-1} = \frac{1}{2} \ln \left| \frac{x-1}{x+1} \right| + C$

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